TEST CODE **01238020**

# FORM TP 2004020

JANUARY 2004

CARIBBEAN EXAMINATIONS COUNCIL

## SECONDARY EDUCATION CERTIFICATE EXAMINATION

## PHYSICS

### Paper 02 – General Proficiency

*$1\frac{1}{2}$ hours*

**READ THE FOLLOWING DIRECTIONS CAREFULLY**

1. You MUST use this answer booklet when responding to the questions. For each question, write your answer in the space provided and return the answer booklet at the end of the examination.
2. ALL WORKING MUST BE SHOWN in this booklet, since marks will be awarded for correct steps in calculations.
3. Attempt ALL questions.
4. The use of non-programmable calculators is allowed.
5. Mathematical tables are provided.

**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO**

Copyright © 2002 Caribbean Examinations Council.
All rights reserved.

1. **You are to spend no more than $\frac{1}{2}$ hour on this question.**

The time for 20 oscillations of a simple pendulum was measured as its length, $l$, was varied. The results are tabulated below.

| **Time for 20 oscillations *t/s*** | 18.8 | 24.4 | 31.0 | 36.4 | 40.0 | 43.2 |
|---|---|---|---|---|---|---|
| **Length *l/m*** | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 1.2 |
| **Period *T/s*** | | | | | | |
| $\sqrt{l}/\mathrm{m}^{1/2}$ | | | | | | |

**Table 1**

(a) (i) Complete Table 1 above. **(4 marks)**

(ii) Explain why it is better to measure twenty timed oscillations instead of one oscillation.

_____

_____

**(1 mark )**

(b) Use the data in the table to plot a graph of $\sqrt{l}$ against T. **(9 marks)**

(c) Find the slope, $S$, of your graph.

_____

_____

_____

**(5 marks)**

(d) Given that $\frac{\sqrt{g}}{6.28} = S$, where $g$ is the acceleration due to gravity, find $g$.

_____

_____

_____

**(2 marks)**

GO ON TO THE NEXT PAGE

GO ON TO THE NEXT PAGE

(e) The mass of the bob is 130 g and the length of the pendulum is 1.2 m. The maximum vertical displacement of the bob above its equilibrium position is 1.8 cm.

(i) Calculate its potential energy at this point.

**(3 marks)**

(ii) a) On the diagram of a simple pendulum shown in Figure 1, indicate, by drawing arrows, the forces acting on the bob. **(2 marks)**

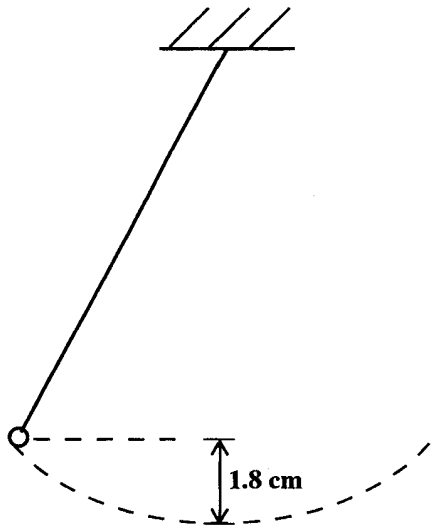


**Figure 1**

b) Name the forces by labelling them on the diagram in Figure 1.

**(2 marks)**

(f) Find the value of the forces on the bob when it is in the equilibrium position.

**(3 marks)**

**Total 31 marks**

**2.** (a) Solar radiation at the earth's surface consists mainly of electromagnetic waves in the wavelength range 0.3 μm to 3.0 μm. Name the THREE types of electromagnetic radiation whose wavelengths lie within this range.

_____

_____

**(3 marks)**

(b) Name ONE type of electromagnetic radiation whose wavelength is

(i) less than 0.3 μm _____

**(1 mark )**

(ii) greater than 3.0 μm. _____

**(1 mark )**

(c) Name ONE property which these three types of radiation have in common.

_____

**(1 mark )**

(d) Calculate the frequency of electromagnetic radiation of wavelength 1.5 μm.

_____

_____

_____

_____

(Velocity of electromagnetic waves in vacuum = $3 \times 10^8$ m/s)

**(3 marks)**

(e) A monochromatic (single wavelength) beam of light enters a glass block whose refractive index is 1.5.

Calculate the velocity of light in the block.

_____

_____

_____

_____

**(3 marks)**

**Total 12 marks**

GO ON TO THE NEXT PAGE

**3.** (a) (i) Draw a circuit diagram showing two resistors $R_1$ and $R_2$ in series with a battery.

**(1 mark )**

(ii) Write down an expression for the equivalent resistance of this combination.

_____

_____

**(1 mark )**

(b) (i) Draw a circuit diagram showing two resistors $R_1$ and $R_2$ in parallel with a battery.

**(1 mark )**

(ii) Write down an expression for the equivalent resistance of this combination.

_____

_____

**(1 mark )**

(c) (i) A set of Christmas lights, consisting of 100 identical bulbs rated at 1.0 W arranged in series, is connected to the 110 V mains. Calculate the voltage across EACH bulb.

**(2 marks)**

(ii) If all the bulbs are ON at the same time, calculate the current through EACH bulb.

**(4 marks)**

(iii) Calculate the resistance of one of these bulbs.

**(3 marks)**

(d) A bimetallic strip may be used to make the lights switch on and off repeatedly. A schematic of such an arrangement is shown in Figure 2.

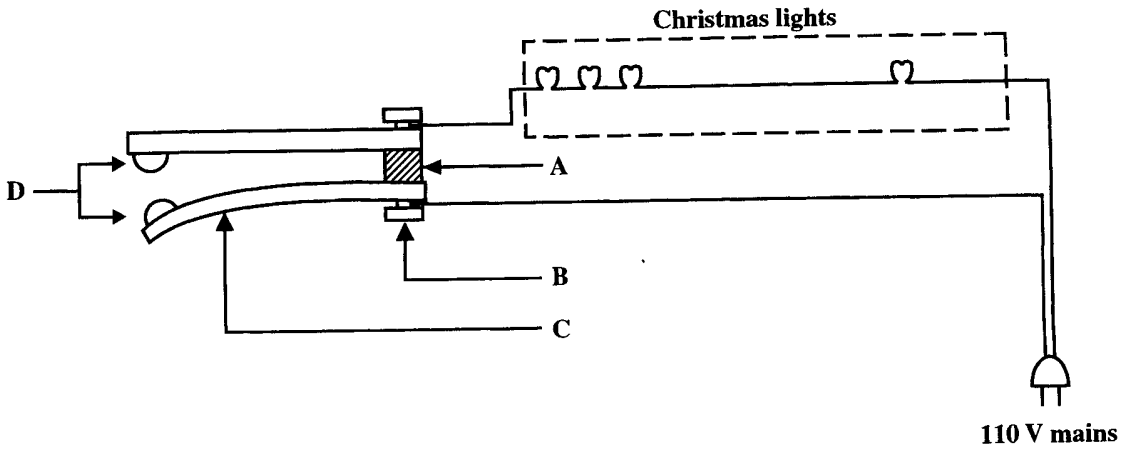


**Figure 2**

(i) Identify the components labelled A, B, C and D.

A _____

B _____

C _____

D _____

**(4 marks)**

(ii) Name TWO metals that could be used to make a bimetallic strip.

_____

_____

**(2 marks)**

**Total 19 marks**

**4.** (a) State Newton's Third Law of Motion.

_____

_____

_____

**(2 marks)**

GO ON TO THE NEXT PAGE

(b) Define the moment of a force and state its SI unit.

**(3 marks)**

(c) Figure 3 shows a wheeled suitcase, of weight $W$, supported in the stationary position shown by applying a vertical force $Y$ at the towing handle.

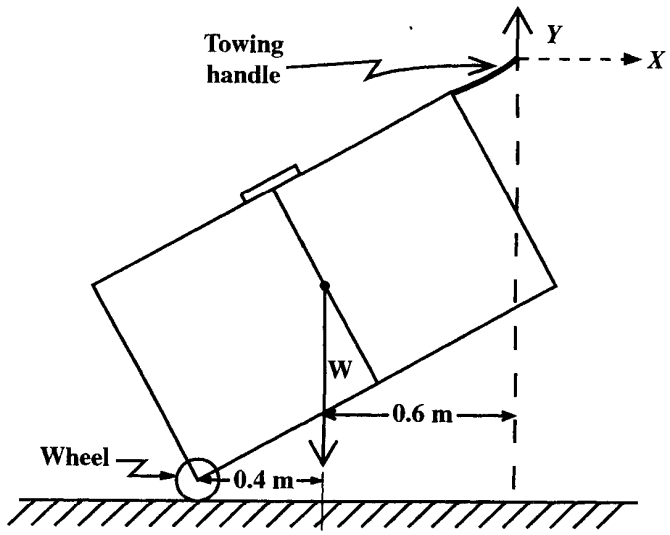


**Figure 3**

The mass of the suitcase is 25 kg. By taking moments about the wheel, calculate the magnitude of the force $Y$.

**(6 marks)**

(d) An additional horizontal force $X$ is applied at the towing handle so that the suitcase begins to move in a horizontal direction.

Deduce the relationship between the force $X$ and the horizontal frictional force at the point of contact with the ground when the suitcase

(i) is moving at constant speed

_____

**(1 mark )**

(ii) is accelerating.

_____

**(1 mark )**

**Total 13 marks**

**5.** (a) Define specific heat capacity and state the relationship between heat capacity and specific heat capacity.

_____

_____

_____

_____

**(3 marks)**

(b) 1.5 kg of water initially at 19° C is heated to 45° C. Calculate the amount of heat energy supplied to the water.

_____

_____

_____

_____

(Specific heat capacity of water = 4200 J $kg^{-1}$ $°C^{-1}$)

**(4 marks)**

(c) Define specific latent heat of vaporisation.

_____

_____

**(2 marks)**

(d) An electric kettle rated at 2.5 kW contains 1.4 kg of water. The kettle is switched on and brings the water to boil. Calculate the time taken to convert the entire amount of water in the kettle to steam after it begins to boil.

_____

_____

_____

_____

_____

_____

(Specific latent heat of vaporisation = $2.26 \times 10^6$ J/kg)

**(6 marks)**

**Total 15 marks**

**END OF TEST**